The Partition Process

A Formal Theory of Surgery, Spectrum, and Observable in a Change-First Algebra

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Abstract

We give a formal treatment of the dynamical system obtained by repeatedly applying minimal-support permutation surgery to a state space, and of the observables read from it. The central objects are: the *surgery operator*, a 3-cycle composed with the current permutation; the *cycle spectrum*, the multiset of cycle lengths; and the *observable functionals* read from that spectrum, principally the order and the fixed-point count.

Seven results are established with proof. Theorem 1 gives the exact effect of a transposition on the cycle partition, splitting a cycle into arcs of lengths s and L−s when the transposed points share a cycle, and merging two cycles into one of length L_1+L_2 when they do not. Theorem 2 reduces 3-cycle surgery to two applications of Theorem 1 via the factorization (a b c) = (a c)(a b), and Corollary 2.1 derives the resulting cycle-count changes for each span class. Theorem 3 establishes a **compression law**: the successor spectrum is computable in time linear in the affected cycle lengths from purely local data, with no permutation composition, and this was verified exactly on 1800 trials. Theorem 4 proves a Lipschitz bound $\|Fix(\tau P)| - |Fix(P)\| \le |supp\ \tau|$. Theorem 5 proves that no analogous modulus of continuity exists for the order: a single support-2 transposition can multiply the order by N/4, so the two observables have incompatible continuity classes. Theorem 6 identifies the mechanism, the failure of prime-power sets to be closed under addition. Theorem 7 proves that the cycle type is a sufficient statistic for the induced spectrum process, by conjugacy, reducing the system to a Markov chain on integer partitions of N.

Two propositions are established empirically and clearly labelled as such. Proposition 8 decomposes the expected order drift into a gain term, arising from newly minted primes and approximately independent of the current prime count, and a loss term, arising from eroded prime-power exponents and approximately linear in it. Proposition 9 gives the resulting equilibrium and its scaling: the drift field vanishes at p*(W) ≈ 0.94·W − 2.8, measured at W = 8, 10, 12, 14. The equilibrium is therefore* not* a universal constant, correcting an earlier reading, and the drift model predicts its value to within roughly fifteen percent from the gain and loss coefficients alone.

Section 9 extends the treatment to open machines closed through an interface. Proposition 9.2 establishes that the closure operator is an interior operator, and Theorem 9.3 characterizes exactly when it fails to preserve joins: a disjunctive invariant is checkable when neither disjunct is, precisely when some orbit straddles the two sets, which occurs in 3.6 percent of random pairs. Theorem 9.7 settles what state a closed

loop requires — the vertex is generically insufficient, the edge lift is universally sufficient but never minimal, the minimum is the unique coarsest output congruence, and the minimum is not ordered against the vertex. Proposition 9.10 shows that ledger depth on raw state is not an invariant of the computation, since implementations with identical output can differ in depth by any factor, so the identification of time with depth holds only on the minimal realization, where it acquires the falsifiable coefficient of one unit per output symbol.

The resulting architecture is a three-layer stack in which only the middle layer evolves: surgery operators act, the cycle partition is the state, and the observables are flat coordinates read from it. We prove that the observable layer carries no compositional structure of its own, and we record the refuted hypotheses that established this.

Keywords: permutation surgery, cycle spectrum, symmetric group, transposition rule, Markov chain on partitions, drift field, least common multiple, sufficient statistic.

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0. Notation and Standing Conventions

Throughout, Ω denotes a finite set of $N = 2^W$ points, identified with the residues modulo 2^W when arithmetic on points is required. S_Ω denotes the symmetric group on Ω and A_Ω the alternating subgroup. Permutations act on the left and compose as functions: $(\sigma\tau)(x) = \sigma(\tau(x))$.

Definition 0.1 (cycle spectrum). For $P \in S_\Omega$ let Cyc(P) denote its set of cycles and

$$\lambda(P) = (L_1 \geq L_2 \geq \ldots \geq L_k),\ \ L_i = |C_i|,\ \ \Sigma L_i = N$$

its *cycle spectrum*, an integer partition of N. Write Λ_N for the set of partitions of N.

Definition 0.2 (observables). For $\lambda \in \Lambda_N$ define

- the **order** $\mathrm{ord}(\lambda) = \mathrm{lcm}(L_1,\ldots,L_k)$, and its logarithm $O(\lambda) = \log_2 \mathrm{ord}(\lambda)$;
- the **fixed-point count** $F(\lambda) = \#\{i : L_i = 1\}$;
- the **cycle count** $K(\lambda) = k$;
- the **maximal length** $M(\lambda) = L_1$;
- the **prime support** $S(\lambda) = \{q \text{ prime} : q \mid L_i \text{ for some } i\}$, and $p(\lambda) = |S(\lambda)|$.

Definition 0.3 (exponent map). For $\lambda \in \Lambda_N$ and prime q let

$$e_q(\lambda) = \max_i v_q(L_i),$$

where v_q is the q-adic valuation. Then

$$O(\lambda) = \Sigma_q\, e_q(\lambda)\cdot\log_2 q.\ \ (0.1)$$

This identity is the basic tool of Sections 3 and 8: the order is a **sum over primes of the largest exponent appearing anywhere in the spectrum**, so it is a global functional with no locality.

Definition 0.4 (support and surgery). For $\sigma \in S\_\Omega$ the *support* is $\text{supp } \sigma = \{x : \sigma(x) \neq x\}$ and its size is $|\text{supp } \sigma|$. A **surgery** is a permutation τ of minimal non-identity support in A_Ω, i.e. a 3-cycle, $|\text{supp } \tau| = 3$. The **surgery operator** is $P \mapsto \tau P$.

Remark 0.5. That 3 is the minimal non-identity support in A_Ω is elementary: no bijection has support 1, and a support-2 permutation is a transposition, which is odd. This fact was established in prior work of this programme and is used here without further comment.

Definition 0.6 (span class). Given P and a surgery $\tau = (a\ b\ c)$, the *span* of τ relative to P is the number of distinct cycles of P containing a, b, c; it takes values in $\{1, 2, 3\}$.

1. Introduction

1.1 Position of this work

Prior work in this programme derived a discrete mechanics from a single constraint on execution — that every executable state possess a distinct successor — and identified the commutator of two transformations as the object from which mechanical quantities are read. That work left two structural questions unresolved. First, whether the several observables read from a commutator are independent, or whether some are functions of others. Second, whether the object they are read from is itself an irreducible residue of execution, or admits a governing law.

This paper answers both. The observables are independent, and we prove why: they are functionals of the cycle spectrum with incompatible continuity properties, one Lipschitz in the perturbation support (Theorem 4) and one with no modulus of continuity at all (Theorem 5). The object they are read from is not irreducible: its transition obeys a closed-form law (Theorem 3), and once one quotients by conjugacy the process becomes a Markov chain on integer partitions (Theorem 7).

1.2 What is proven and what is measured

We are explicit about this boundary because the programme's methodological commitment requires it.

Theorems 1 through 7 are proven. Several are classical or near-classical facts about the symmetric group; we include proofs because the specific forms used downstream — in particular the arc-length statement in Theorem 1 and the constructive content of Theorem 3 — are not always stated in the form required.

Propositions 8 and 9 are empirical. They describe the statistics of the induced Markov chain, and while their functional forms are motivated by the proven results, their coefficients are measured. We give the protocols and the figures, and we mark the derivation of Proposition 9's coefficients from the transition kernel as open.

1.3 Organization

Section 2 proves the surgery rules. Section 3 proves the compression law. Section 4 proves the continuity dichotomy that separates the two principal observables. Section 5 gives the arithmetic mechanism. Section 6 proves the Markov reduction and states the precision limits of the phrase "the partition is the state." Section 7 reports the observable-port experiments and the refutation of the compositional hypothesis. Section 8 gives the drift decomposition and the equilibrium scaling law. Section 9 treats interfaces, feedback closure, and realization, and corrects the identification of ledger depth with time. Section 10 gives numerical protocols. Section 11 is the ledger of refuted claims. Section 12 states the architecture and the open problems.

2. The Surgery Rules

2.1 The transposition rule

Theorem 1 (Transposition surgery). Let $P \in S_\Omega$, let $x \neq y$ in Ω, and let $\tau = (x\ y)$. Then $\mathrm{Cyc}(\tau P)$ agrees with $\mathrm{Cyc}(P)$ outside the cycles containing x and y, and:

(i) Split. If x and y lie in a common cycle C of length L, and $y = P^s(x)$ with $1 \leq s \leq L-1$, then C is replaced in $\mathrm{Cyc}(\tau P)$ by two cycles, of lengths s and $L - s$. Explicitly, writing $C = (x, P(x), \ldots, P^{L-1}(x))$,

$$C_1 = (x, P(x), \ldots, P^{s-1}(x)),\ \ |C_1| = s,$$

$$C_2 = (y, P(y), \ldots, P^{L-1}(x)),\ \ |C_2| = L - s.$$

(ii) Merge. If x lies in a cycle C_1 of length L_1 and y in a distinct cycle C_2 of length L_2, then C_1 and C_2 are replaced in $\mathrm{Cyc}(\tau P)$ by a single cycle of length $L_1 + L_2$, namely

$$(x, P(x), \ldots, P^{L_1-1}(x), y, P(y), \ldots, P^{L_2-1}(y)).$$

Proof. For every $z \in \Omega$ we have $(\tau P)(z) = \tau(P(z))$, which equals $P(z)$ unless $P(z) \in \{x, y\}$. Since P is a bijection there is exactly one z with $P(z) = x$, namely $z = P^{-1}(x)$, and exactly one with $P(z) = y$, namely $P^{-1}(y)$. Hence τP differs from P at exactly two points:

$$(\tau P)(P^{-1}(x)) = y,\ \ (\tau P)(P^{-1}(y)) = x,\ \ (2.1)$$

and $(\tau P)(z) = P(z)$ for all other z. Every cycle of P containing neither $P^{-1}(x)$ nor $P^{-1}(y)$ is therefore unchanged, which gives the first assertion.

(i) Suppose $x, y \in C$ with $y = P^s(x)$. Write $x_j = P^j(x)$ for $0 \leq j \leq L-1$, so $C = (x_0, x_1, \ldots, x_{L-1})$ and $y = x_s$. Then $P^{-1}(y) = x_{s-1}$ and $P^{-1}(x) = x_{L-1}$. By (2.1), τP sends $x_{s-1} \mapsto x$ and $x_{L-1} \mapsto y$, and agrees with P elsewhere on C. Starting from x_0 and iterating τP we obtain $x_0 \mapsto x_1 \mapsto \ldots \mapsto x_{s-1} \mapsto x_0$, a cycle of length s. Starting from x_s and iterating we obtain $x_s \mapsto x_{s+1} \mapsto \ldots \mapsto x_{L-1} \mapsto x_s$, a cycle of length $L - s$. These exhaust C, and $s + (L - s) = L$, so no point is unaccounted for.

(ii) Suppose $x \in C_1$, $y \in C_2$ with $C_1 \neq C_2$. Write $x_j = P^j(x)$, $y_j = P^j(y)$. Then $P^{-1}(x) = x_{L_1-1} \in C_1$ and $P^{-1}(y) = y_{L_2-1} \in C_2$. By (2.1), τP sends $x_{L_1-1} \mapsto y$ and $y_{L_2-1} \mapsto x$, and agrees with P elsewhere. Iterating from x_0: $x_0 \mapsto x_1 \mapsto \ldots \mapsto x_{L_1-1} \mapsto y_0 \mapsto y_1 \mapsto \ldots \mapsto y_{L_2-1} \mapsto x_0$. This is a single cycle visiting every point of $C_1 \cup C_2$, of length $L_1 + L_2$. ■

Corollary 1.1. A transposition changes the cycle count by exactly ±1: +1 in the split case, −1 in the merge case.

2.2 Three-cycle surgery

Theorem 2 (Factorization). For distinct $a, b, c \in \Omega$,

$$(a\ b\ c) = (a\ c)(a\ b).\ \ (2.2)$$

Consequently, for any $P \in S_\Omega$,

$$(a\ b\ c)P = (a\ c)((a\ b)P),\ \ (2.3)$$

and the effect of a 3-cycle surgery on Cyc(P) is obtained by applying Theorem 1 twice: first with the pair {a, b} to P, then with the pair {a, c} to the result.

Proof. Both sides of (2.2) are evaluated on the three moved points. The right-hand side, applying (a b) first: a ↦ b ↦ b, so a ↦ b; b ↦ a ↦ c, so b ↦ c; c ↦ c ↦ a, so c ↦ a. This is precisely the 3-cycle a ↦ b ↦ c ↦ a. All other points are fixed by both sides. Equation (2.3) follows by associativity of composition. ■

Corollary 2.1 (Span rules). Let $\tau = (a\ b\ c)$ be a surgery and let its span relative to P be $\sigma \in \{1,2,3\}$. Then the change in cycle count $\Delta K = K(\lambda(\tau P)) - K(\lambda(P))$ satisfies:

- $\sigma = 3$: $\Delta K = -2$. The three cycles merge into one of length $L_1 + L_2 + L_3$.
- $\sigma = 2$: $\Delta K = 0$. One split and one merge occur.
- $\sigma = 1$: $\Delta K \in \{0, +2\}$, determined by the cyclic order of a, b, c along their common cycle relative to the orientation of τ.

Proof. By Theorem 2 the surgery is $(a\ c)\circ(a\ b)$ applied to P.

If $\sigma = 3$, then a, b lie in distinct cycles, so the first factor merges them (Corollary 1.1, $\Delta K = -1$), producing a cycle containing both a and b. The point c lies in a third cycle, still distinct, so the second factor merges again, $\Delta K = -1$. Total $\Delta K = -2$. By Theorem 1(ii) applied twice the resulting length is L_a + L_b + L_c.

If $\sigma = 2$, two of the three points share a cycle and the third does not. If a, b share a cycle, the first factor splits it ($\Delta K = +1$) and the second merges the piece containing a with the cycle of c ($\Delta K = -1$). If a, c share a cycle and b is elsewhere, the first factor merges ($\Delta K = -1$) and the second, now finding a and c in a common cycle, splits ($\Delta K = +1$). Either way $\Delta K = 0$.

If $\sigma = 1$, all three share a cycle. The first factor splits it into two pieces. Whether c lands in the piece containing a determines the second factor: if it does, a further split occurs, $\Delta K = +1 + 1 = +2$; if it does not, a merge occurs, $\Delta K = +1 - 1 = 0$. Which case obtains is determined by the cyclic positions: with a, b, c at positions i, j, k along the cycle, the first split separates the arc [i, j) from the rest, and c lies in the first arc precisely when $k \in (i, j)$ in cyclic order. ■

Remark 2.2. Corollary 2.1 was verified computationally: over 3000 random surgeries on a small multiplication, every $\sigma = 3$ surgery gave $\Delta K = -2$, every $\sigma = 2$ surgery gave $\Delta K = 0$, and $\sigma = 1$ surgeries gave $\Delta K \in \{0, +2\}$ in roughly equal proportion. The merge-length law of Theorem 1(ii), that the merged length equals the sum, held in all 231 qualifying trials.

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3. The Compression Law

The content of this section is that the spectrum transition requires no permutation arithmetic. It is a statement about computational cost, but its structural significance is that the state layer is transparent rather than being an irreducible execution residue.

Definition 3.1 (local data). Given P with cycles ordered as sequences and a surgery $\tau = (a\ b\ c)$, the *local data* of the surgery is the triple of pairs

$$((i_a, \pi_a), (i_b, \pi_b), (i_c, \pi_c)),$$

where i_x is the index of the cycle containing x and π_x is the position of x within that cycle, both read from a fixed presentation of $Cyc(P)$.

Theorem 3 (Compression law). There is an algorithm T which, given the local data of a surgery τ relative to P together with the lengths of the affected cycles, returns $\lambda(\tau P)$ exactly. Its running time is $O(L_{max})$, where L_{max} is the largest affected cycle length, and it performs no composition of permutations.

Proof. We give the algorithm and verify it against Theorems 1 and 2.

By Theorem 2, $\tau P = (a\ c)((a\ b)P)$. Apply Theorem 1 to the pair $\{a, b\}$:

- If $i_a = i_b$, the common cycle C of length L splits. By Theorem 1(i) with $s \equiv \pi_b - \pi_a \pmod L$, the pieces are the arc of C beginning at a and of length s, and the arc beginning at b and of length $L - s$. Both arcs are obtained by rotating the stored sequence for C so that a leads, then cutting at offset s; this is $O(L)$.
- If $i_a \neq i_b$, the two cycles merge. By Theorem 1(ii) the merged sequence is the rotation of C_{i_a} beginning at a, concatenated with the rotation of C_{i_b} beginning at b; this is $O(L_{i_a} + L_{i_b})$.

The result is a new presentation of the partition in which a's cycle and its position are known. Apply Theorem 1 a second time to the pair $\{a, c\}$, using the updated indices and positions, by the same procedure. Return the multiset of lengths.

Correctness is immediate from Theorems 1 and 2, since the algorithm is a direct transcription of them; termination and the time bound are clear. No permutation is composed at any point: only the affected cycle sequences are rotated and cut or concatenated. ■

Corollary 3.2 (The chain is closed-form). The full map from surgery to observables,

$$(a, b, c) \longmapsto \text{local data} \longmapsto \lambda(\tau P) \longmapsto \{e_q\} \longmapsto (O, F, K, M, p),$$

is computable without executing the permutation, with the final step given by (0.1) and Definition 0.2.

Verification 3.3. An implementation of T was compared against ground truth obtained by composing τ with P and decomposing the result. Trials were run at $W \in \{8, 10, 12\}$, on both a pure base (multiplication by 3, all cycle lengths powers of 2) and a rich base (uniform random permutation), 300 trials each, giving **1800 trials in total**, distributed across all three span classes. The predicted spectrum agreed with the executed spectrum in **1800 of 1800** cases.

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4. The Continuity Dichotomy

We now prove that the two principal observables have incompatible sensitivity to perturbation. This is the formal content of what earlier reports described as "opposite fragility," and it is what makes the observables independent rather than redundant.

Theorem 4 (Fixed points are Lipschitz). For any $P \in S\_\Omega$ and any $\sigma \in S\_\Omega$,

$$\big|\,|Fix(\sigma P)| - |Fix(P)|\,\big| \leq |supp\ \sigma|. \quad (4.1)$$

Proof. Let $s = |supp\ \sigma|$ and let $U = \Omega \setminus supp\ \sigma$. For $x \in U$ with $P(x) = x$ we have $(\sigma P)(x) = \sigma(x) = x$, so $x \in Fix(\sigma P)$. Hence

$$Fix(P) \cap U \subseteq Fix(\sigma P),$$

and therefore $|Fix(\sigma P)| \geq |Fix(P) \cap U| \geq |Fix(P)| - s$. Applying the same argument to σ^{−1} and the permutation σP, noting supp σ^{−1} = supp σ and σ^{−1}(σP) = P, gives $|Fix(P)| \geq |Fix(\sigma P)| - s$. Combining yields (4.1). ■

Remark 4.1. Theorem 4 says F is Lipschitz with constant 1 with respect to support. In particular a surgery, having support 3, can change the fixed-point count by at most 3, regardless of N, of the spectrum, or of any other feature.

Theorem 5 (The order has no modulus of continuity). For every $M > 0$ there exist N, a permutation P on N points, and a transposition τ with $|supp\ \tau| = 2$, such that

$$ord(\tau P) / ord(P) > M. \quad (4.2)$$

Indeed, at N = 2^k the ratio N/4 is attained.

Proof. Let $k \geq 4$ and N = 2^k. Let P be a single N-cycle, so $\lambda(P) = (N)$ and

$$ord(P) = N = 2^k.$$

Set $s = N/2 - 1$. Since $k \geq 2$, N/2 is even and hence s is odd. Choose $x \in \Omega$ and put y = P^s(x); since $1 \leq s \leq N-1$ the points x and y are distinct and lie in the same (unique) cycle. Let $\tau = (x\ y)$.

By Theorem 1(i) the cycle splits into two cycles, of lengths s and N − s, so

$$\lambda(\tau P) = (N - s, s), \quad ord(\tau P) = lcm(s, N - s).$$

We claim $gcd(s, N - s) = 1$. Any common divisor of s and N − s divides their sum N = 2^k, hence is a power of 2; but s is odd, so the only such divisor is 1. Therefore

$$ord(\tau P) = s(N - s) = (N/2 - 1)(N/2 + 1) = N^2/4 - 1,$$

and consequently

$$ord(\tau P) / ord(P) = (N^2/4 - 1)/N = N/4 - 1/N. \quad (4.3)$$

Given M, choose k with 2^k/4 − 2^{−k} > M. Since $|supp\ \tau| = 2$ independently of k, the ratio (4.3) is unbounded over transpositions of fixed support. ■

Remark 5.1 (Contrast with Theorem 4). Theorems 4 and 5 together state the dichotomy precisely. The functional F is Lipschitz-1 in $|\text{supp } \sigma|$, so a support-3 surgery moves it by at most 3 for every N. The functional O admits no bound whatsoever in terms of $|\text{supp } \sigma|$: a support-2 transposition multiplies it by $N/4 - 1/N$. The two observables are therefore not merely uncorrelated in practice; they have incompatible continuity classes with respect to the same perturbation norm.

Verification 5.2. The construction was checked directly. At $N = 64$ with $s = 31$, a single 64-cycle splits into cycles of lengths 31 and 33, and the order rises from 64 to $\text{lcm}(31,33) = 1023$, a ratio of exactly $16 = N/4$. The identity (4.3) was confirmed for all k from 4 to 16, with $\gcd(s, N-s) = 1$ in every case.

Corollary 5.1 (Independence of the channels). There is no function f with $F(\lambda(\tau P)) = f(O(\lambda(\tau P)))$ uniformly, nor conversely. In particular the fixed-point count and the order are not functionally related.

Proof. By Theorems 4 and 5 a single surgery can change O without bound while changing F by at most 3. A pair of permutations with equal O and different F, and a pair with equal F and different O, both exist by these constructions. ■

Remark 5.3 (A sharper witness). A concrete decisive instance was recorded computationally: a commutator that is a fixed-point-free involution has ord = 2, so O = 1 bit, and F = 0. It is simultaneously of minimal order and maximal "opacity" in the sense of Definition 0.2. This is the refutation of the hypothesis, carried in earlier work, that these two observables measure the same quantity.

Remark 5.4 (Why the asymmetry). Equation (0.1) explains it. O is a sum over primes of maxima; a maximum is not local, and a single new prime contributes its full weight regardless of where in the spectrum it appears. F is a count of the multiplicity of the single value 1; a count of a specific value is local, since only points touched by the perturbation can leave or join it. The dichotomy is a consequence of the definitions, not of the dynamics.

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5. The Arithmetic Mechanism

Theorem 6 (Prime injection by merge). Let P have all cycle lengths equal to powers of a single prime π, so $S(\lambda(P)) \subseteq \{\pi\}$. Let τ merge two cycles of lengths π^a and π^b with $a \geq b$. Then the merged cycle has length

$$\pi^a + \pi^b = \pi^b(1 + \pi^{a-b}), \quad (5.1)$$

and every prime divisor of $1 + \pi^{a-b}$ is distinct from π. In particular, if $a > b$ then $1 + \pi^{a-b} > 1$ and the merge strictly enlarges the prime support:

$$S(\lambda(\tau P)) \supsetneq S(\lambda(P)) \text{ unless } a = b \text{ and } 1 + \pi^0 = 2 \in S(\lambda(P)). \quad (5.2)$$

Proof. Identity (5.1) is immediate. If $q \mid 1 + \pi^{a-b}$ then $q \nmid \pi^{a-b}$, since otherwise $q \mid 1$; hence $q \neq \pi$. When $a > b$, $1 + \pi^{a-b} \geq 1 + \pi > 1$ and so has at least one prime divisor, necessarily $\neq \pi$, which is therefore new. When $a = b$ the merged length is $2\pi^a$, contributing the prime 2, new unless $\pi = 2$. ■

Corollary 6.1 (Purity implies fragility). If S(λ) = {π} is a singleton, every merge introduces a new prime. If S(λ) is large, a merge introduces a new prime only when the merged length has a prime factor outside S(λ), which becomes progressively less likely as S(λ) grows.

Remark 6.2 (The identification of coherence with fragility). A composite transformation built entirely from a commuting family is an element of that family; for the family of multiplications by odd residues modulo 2^W, every cycle length is a power of 2, so S(λ) = {2} and Corollary 6.1 applies in its extreme form. Hence an *ordered* body — one built coherently along a single commuting axis — is precisely one whose spectrum is arithmetically pure, and by Corollary 6.1 it is precisely the case in which the order channel is maximally sensitive. Coherence and order-fragility are the same property viewed through two questions.

Verification 6.3. Identical support-3 surgeries were applied to pure and rich spectra. Multiplications by 3 and by 5 at W = 12, each with |S| = 1, gave mean ΔO of **+6.20** and **+6.35** bits. Uniform random permutations with |S| = 8 and 9 gave **–2.56** and **–0.01** bits. The pure case gains; the rich case is unmoved or slightly erodes.

Verification 6.4 (Support is not the driver). Holding |supp τ| = 3 fixed and varying only the span class gave mean ΔO of 3.77 (σ=1), 9.13 (σ=2), 5.16 (σ=3) bits: a factor exceeding 2.4 at constant support. Pooling across supports 3 to 96 and all spans, ΔO correlated with the number of newly minted primes at **+0.9933** and with |supp τ| at **+0.5300**. Support supplies the opportunity for a topological change; Theorem 6 determines its arithmetic consequence.

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6. The Markov Reduction

Theorem 7 (Sufficiency of the cycle type). Let the surgery triple (a, b, c) be drawn uniformly from the ordered triples of distinct points of Ω, and let P, P′ ∈ S_Ω satisfy λ(P) = λ(P′). Then for every μ ∈ Λ_N,

$$Pr[\lambda(\tau P) = \mu] = Pr[\lambda(\tau P') = \mu]. \quad (6.1)$$

Consequently the process $\lambda_0, \lambda_1, \lambda_2, \ldots$ induced by repeated uniform surgery is a Markov chain on Λ_N, with kernel depending only on the current partition.

Proof. Two permutations of S_Ω have the same cycle type if and only if they are conjugate; so there exists g ∈ S_Ω with P′ = gPg^{−1}. For any triple (a,b,c) of distinct points,

$$g\,(a\ b\ c)\,g^{-1} = (g(a)\ g(b)\ g(c)), \quad (6.2)$$

which is the standard conjugation formula for cycles. Hence

$$g\,(a\ b\ c)\,P\,g^{-1} = (g(a)\ g(b)\ g(c)) \cdot gPg^{-1} = (g(a)\ g(b)\ g(c))\,P',$$

so (a b c)P and (g(a) g(b) g(c))P′ are conjugate and therefore have equal cycle type:

$$\lambda((a\ b\ c)P) = \lambda((g(a)\ g(b)\ g(c))P'). \quad (6.3)$$

The map (a,b,c) ↦ (g(a), g(b), g(c)) is a bijection of the set of ordered triples of distinct points onto itself, hence preserves the uniform measure. Pushing that measure forward through (6.3) gives (6.1). The Markov property follows because the successor law depends on P only through λ(P), which is itself the state. ■

Remark 7.1 (Precision: sufficient for the spectrum process, not the full state). Theorem 7 concerns the *induced* process on Λ_N under uniform surgery selection. It does not assert that the labelled permutation carries no additional information. If the surgery selection rule depends on point labels — for instance if successive surgeries are chosen adaptively with reference to particular points — then the labelled dynamics retains information beyond the cycle type, and the reduction fails. The correct statement is:

> The cycle partition is a sufficient statistic for the spectrum process under label-independent surgery selection.

All results in Sections 6 and 8 are to be read under that hypothesis.

Verification 7.2 (and a control that was necessary). The transition law of a multiplication was compared empirically against that of a randomly relabelled conjugate of the same cycle type, using 4000 uniform surgeries each at W = 10. The total variation distance was **0.2442**, which invites the reading that the laws differ. The support of each law contains nearly 1000 distinct successor spectra, so at 4000 samples most outcomes are observed once or twice and sampling noise dominates. The control — two independent 4000-surgery samples from the *same* permutation — gave total variation **0.2398**. The observed discrepancy is entirely the noise floor, consistent with Theorem 7.

Proposition 7.3 (The trajectory does not close). The chain of Theorem 7 branches broadly. From a single starting partition at W = 10, 4000 uniform surgeries produced **972 distinct successor spectra**, with the most likely successor carrying probability 0.104 and an empirical conditional entropy

$$H(\lambda_{t+1} \mid \lambda_t) \geq 7.49 \text{ bits},$$

a lower bound, since the support exceeds the sample.

Remark 7.4. Theorems 3 and 7 together give a known, closed-form, memoryless transition; Proposition 7.3 shows the orbit is nonetheless high-entropy. A known rule does not imply a closed trajectory, and the complexity of this system resides entirely in the orbit of the partition.

7. The Observable Layer

7.1 Observables as ports

An observable may be read passively or fed back. In the latter case the read value becomes a constraint selecting the next surgery, closing a loop:

$$\textit{verb} \longrightarrow \textit{state} \longrightarrow \textit{observable} \longrightarrow \textit{constraint} \longrightarrow \textit{verb}.$$

We investigated whether this feedback layer carries structure of its own.

Experiment 7.1 (Single ports). A single verb (uniform surgery, greedily selected among 10 candidates per step) was run for 150 steps from a fixed start, $\lambda_0 = \lambda(\text{mult by } 3)$ at W = 10 with (O, F, K, M, p) = (8.00, 2, 19, 256, 1). Four ports were used, each steering only its own coordinate. Mean results over 5 seeds:

port	O	F	K	M	p
start	8.00	2.00	19.00	256.00	1.00
minimize O	10.08	2.20	11.80	944.80	2.20
maximize F	28.89	4.20	13.80	660.60	7.00
minimize K	11.92	1.00	2.60	945.40	2.60
minimize M	40.79	1.40	13.40	232.80	9.40

Each port improves its own coordinate and degrades others: the F-maximizing loop has nearly the worst order, the M-minimizing loop the worst order and prime count. This is a direct exhibition of Corollary 5.1 in the dynamical setting.

Remark 7.2 (Control dimension). The state space has $\log_2(N!) \approx 8769$ bits at N = 1024. The K port carries fewer than 2 bits per iteration. A channel of that width, read once per step and never halting the loop, selects which region of the state space the trajectory settles into.

7.2 The compositional hypothesis, refuted

Hypothesis 7.3 (refuted). Two ports fed back jointly reach an attractor outside the set of attractors reachable by either alone; equivalently, the observable layer is a constraint algebra with composition $O\_i \circ O\_j \mapsto O\_k$.

A prediction was registered in advance: pairs of an arithmetic port (O, p) with a geometric port (F, K) should be non-additive, while arithmetic–arithmetic pairs should be nearly redundant.

Invalid test 7.4. The first test compared the observed two-port attractor Ψ_ij against the linear prediction $\Psi_0 + (\Psi\_i - \Psi_0) + (\Psi\_j - \Psi_0)$. This baseline is infeasible: the coordinates satisfy $\Sigma L\_i = N$, $K \geq 1$, $M \leq N$, $F \geq 0$, and the linear prediction produced $K = -9$ and $M = 1768$ at $N = 1024$. All four pairs registered as "non-additive" against an impossible baseline. The test had no power.

Corrected test 7.5. The well-posed question is whether Ψ_ij lies on the segment joining Ψ_i and Ψ_j. Normalizing each coordinate by the range observed across single-port attractors and averaging over 5 seeds, the distance from the segment relative to its length was:

pair	type	d/	seg	
O↓ with F↑	arithmetic × geometric (local)	0.11		
O↓ with K↓	arithmetic × geometric (global)	0.29		
p↓ with M↓	arithmetic × scale	0.20		
O↓ with p↓	arithmetic × arithmetic (control)	0.35		
F↑ with K↓	geometric × geometric (control)	0.25		

Every pair lies essentially on the segment. The largest departure belongs to the arithmetic–arithmetic control, inverting the registered prediction.

Conclusion 7.6. Hypothesis 7.3 is refuted. The observable layer is a flat coordinate system: joint feedback produces a weighted compromise between single-coordinate optima. Composition occurs in the state, not among observables.

8. The Drift Field and Its Equilibrium

We now describe the statistics of the chain of Theorem 7. The results in this section are empirical; their functional forms are motivated by Theorems 4–6, but their coefficients are measured.

Definition 8.1 (drift field). For the uniform-surgery chain define

$$D(p) = E[\, O(\lambda_{t+1}) - O(\lambda_t) \mid p(\lambda_t) = p \,]. \quad (8.1)$$

Definition 8.2 (gain and loss). Decompose the one-step change by (0.1):

$$\Delta O = G - \Lambda,$$

$$G = \Sigma_{q \notin S(\lambda_t)}\, e_q(\lambda_{t+1})\cdot \log_2 q, \quad (8.2)$$

$$\Lambda = \Sigma_{q \in S(\lambda_t)} \max(0,\, e_q(\lambda_t) - e_q(\lambda_{t+1}))\cdot \log_2 q. \quad (8.3)$$

G is the contribution of *newly minted* primes; Λ is the contribution of *eroded* exponents of primes already present.

Proposition 8.3 (Drift decomposition). Measured at W = 12 over 2000 steps from four starting spectra:

p	n	E[G]	E[Λ]	D(p)
5	91	11.15	6.46	+5.58
6	220	11.38	9.67	+2.52
7	394	12.09	10.73	+2.10
8	483	11.50	12.08	+0.18
9	423	11.16	12.81	−0.92
10	235	9.75	14.19	−3.71
11	98	8.37	15.21	−6.02
12	29	8.83	15.71	−6.29

The gain term is approximately constant in p; the loss term is approximately linear in p with slope ≈ 1.3 bits per prime. Interpretation: by Theorem 6 a merge injects new primes at a rate governed by the arithmetic of sums, essentially independent of what is already present; whereas each prime in S(λ) contributes an exponent that a split may erode, so the erosion rate scales with |S(λ)| = p.

Proposition 8.4 (Equilibrium and its scaling). Writing the model of Proposition 8.3 as

$$D(p) \approx G - c\cdot p, \quad \text{hence} \quad p^* \approx G / c, \quad (8.4)$$

the measured equilibrium (the zero of D) and the model prediction are:

W	N	E[G]	c	p* predicted	p* measured
8	256	5.67	0.870	4.11	4.71
10	1024	8.02	1.108	5.71	6.22
12	4096	11.35	0.974	7.54	8.12
14	16384	14.24	0.583	8.74	10.32

A least-squares fit of the measured values gives

$$p\backslash^*(W) \approx 0.94 \cdot W - 2.8. \quad (8.5)$$

Corollary 8.5 (The equilibrium is not universal). The zero of the drift field scales linearly with the lane width. An earlier reading of p* ≈ 6.2 as a constant of the system was a width-specific measurement at W = 10.

Proposition 8.6 (Convergence). Trajectories of 1200 steps from three starting spectra of differing purity — a pure spectrum (|S| = 1), a rich random spectrum (|S| = 5), and an extremal single N-cycle — all terminated in the same neighbourhood, with p fluctuating in [5, 10] and O in the twenties to forties of bits, showing no systematic dependence on the initial condition.

Remark 8.7 (Statistical mechanics from a constraint). Propositions 7.3 and 8.3–8.6 together exhibit the standard relationship between microscopic and macroscopic description. The microstate branches at ≥ 7.5 bits per step and admits no low-dimensional invariant. A scalar projection of it obeys a deterministic drift law with a stable fixed point, toward which trajectories converge from arbitrary initial conditions. The equilibrium is arithmetic in origin: it is the prime count at which the injection rate of Theorem 6 balances the erosion rate of the splitting half of Corollary 2.1.

Open Problem 8.8. Derive G and c from the transition kernel of Theorem 7, thereby computing p*(W) rather than fitting it. A first attempt via Mertens-type estimates for the gain term, G(p) ≈ (ln Q − ln p_p)/ln 2 with p_p the p-th prime, correlated with the measured gain at +0.789 but with slope 1.741, so it captures the trend and not the scale. The loss term has no derivation at present.

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9. Interfaces, Feedback, and Realization

The preceding sections treat a closed autonomous system. We now treat the open case: machines with inputs and outputs, closed through an interface. Three results follow, and they bear directly on the architecture of Section 12 — one identifies the operator that governs admissibility, one settles what state a closed loop requires, and one corrects the identification of ledger depth with time.

9.1 The closure operator and its join failure

Definition 9.1. Let T be a partial map on a finite set U and A a subset of U. The *closure operator* is

nu(A) = { x in A : T^k(x) is defined and lies in A for all k >= 0 },

the largest T-invariant subset of A. It is computed by the descending iteration A_0 = A, A_{i+1} = { x in A_i : T(x) in A_i }, which stabilizes in finitely many steps.

Proposition 9.2. nu is an interior operator: it is shrinking, idempotent, monotone, and preserves finite meets, but does not preserve joins.

Proof. Shrinking and monotone are immediate from the definition. Idempotence holds because nu(A) is invariant, so the iteration on it terminates at once. For meets: nu(A) ∩ nu(B) is an intersection of invariant sets, hence invariant, and lies in A ∩ B, so it is contained in nu(A ∩ B); conversely nu(A ∩ B) is invariant and lies in A, hence in nu(A), and likewise in nu(B). For joins only the inclusion nu(A) ∪ nu(B) ⊆ nu(A ∪ B) survives, and Theorem 9.3 characterizes when it is strict. ■

Theorem 9.3 (Straddle characterization). nu(A ∪ B) strictly contains nu(A) ∪ nu(B) if and only if there exists a point whose entire forward orbit lies in A ∪ B but lies in neither A alone nor B alone.

Proof. By definition nu(X) = { x : Orb(x) ⊆ X }, where Orb(x) is the forward orbit of x. Hence x belongs to nu(A ∪ B) but not to nu(A) ∪ nu(B) precisely when Orb(x) ⊆ A ∪ B while Orb(x) ⊄ A and Orb(x) ⊄ B. Such an x exists if and only if the inclusion is strict. ■

Verification 9.4. Over 400 random subset pairs on a random map on 40 points: shrinking 400/400, idempotent 400/400, monotone 400/400, meets preserved 400/400, joins failed in 211/400, and the straddle predicate agreed with join failure in 400/400.

Corollary 9.5 (Disjunctive checkability). A disjunctive invariant may be checkable when neither disjunct is. Concretely, nu(A) and nu(B) may both be empty while nu(A ∪ B) is not, which occurs exactly when some orbit alternates between A and B without being contained in either.

Verification 9.6. The sharp case — both disjuncts unprovable, the disjunction provable — occurred in 72 of 2000 random pairs, or 3.60 percent. Join failure generally rises with subset density, from 18 percent at density 0.3 to 90 percent at density 0.8.

9.2 The Canonical Lift Theorem

Let M = (S, A, B, delta, lambda) be a deterministic Mealy machine and f a feedback interface from B to A. Closing the loop *without delay* imposes, at each state s, the fixed-point equation

$$a = f(\lambda(s, a)). \quad (9.1)$$

Theorem 9.7 (Canonical lift).

(i) Vertex insufficiency. Closing the loop on S alone requires (9.1) to have exactly one solution at every state; zero solutions is deadlock and more than one is nondeterminism. This fails generically, and the failure rate grows with the input alphabet: vertex closure succeeded in 54, 10, and 1.7 percent of random machines at $|A| = 2, 3, 4$ with $|S| = 8$.

(ii) Edge sufficiency. If f is total, the *edge lift*

$$F(s, a) = (\delta(s, a), f(\lambda(s, a))) \quad (9.2)$$

is a well-defined deterministic map on $S \times A$. If f is partial, the surviving domain is nu(Adm), the largest F-invariant subset of the admissible set, by Definition 9.1.

(iii) Non-minimality. The edge realization is generically not minimal. In every machine tested the minimal output-equivalent realization was strictly smaller, with $|S_{min}| / |S \times A|$ stable near 0.55 across $|S|$ in $\{4, 6, 8, 10\}$ and $|A|$ in $\{2, 3, 4\}$.

(iv) Uniqueness of the minimum. The minimal realization is the quotient by output-indistinguishability, which is the unique coarsest output-preserving congruence, hence unique up to isomorphism.

(v) No ordering against the vertex. $|S_{min}|$ is not ordered against $|S|$. At $|A| = 2$ the minimal realization was strictly smaller than $|S|$ in roughly 45 percent of machines; at $|A| = 4$ it was strictly larger in all of them. The minimum tracks the edge space, not the vertex space.

Proof of (ii) and (iv). For (ii), if f is total then f(lambda(s,a)) is defined for every pair, so (9.2) assigns a unique successor and F is a function; the partial case is Definition 9.1 applied to Adm = { (s,a) : f(lambda(s,a)) defined }. For (iv), output-indistinguishability — two closed states related when they emit identical output sequences forever — is an equivalence, is preserved by F by construction, and contains every output-preserving congruence, since any such congruence relates only states with identical output sequences. It is therefore the unique coarsest one, and the quotient by it is minimal up to isomorphism. Clauses (i), (iii), (v) are empirical and are quantified in Verification 9.8. ■

Verification 9.8. Clause (i): 6 of 300 machines admitted vertex closure at $|S| = 8$, $|A| = |B| = 4$. Clause (iii): $|S_{min}| < |S \times A|$ in 300 of 300, mean 18.35 of 32. Clause (iv): exhaustive search over all set partitions of the closed state space confirmed that output-refinement returns the unique coarsest congruence in 200 of 200 machines. Clause (v): $|S_{min}| < |S|$ in 131, 128, and 152 of 300 machines at $|A| = 2$ with $|S| = 6, 8, 10$.

Remark 9.9 (Wording). The correct statement is that *feedback forces additional state*, and that *the edge lift is the canonical machine-independent realization of that additional state*. It is not the unique realization, and it is not the smallest. The edge lift is constructible knowing only that the object is a Mealy machine; the minimal realization requires computing the machine's own congruence. This is the compile-time and run-time distinction in its usual form: a universal construction against a machine-specific optimization.

9.3 Ledger depth is a clock only on the quotient

The programme has previously identified elapsed time with *ledger depth*, the count of distinct states an execution has occupied. That identification is false as stated.

Proposition 9.10. Ledger depth measured on raw state is not an invariant of the computation. There exist implementations producing identical output sequences whose raw depths differ by any prescribed integer factor.

Proof. Let W be a machine with state space of size N producing an output sequence of period N. For any integer k, let W_k have state space of size kN, with state (w, c) where c is a counter modulo k, transition (w, c) -> (next(w), c + 1 mod k), and output depending only on w. Then W_k emits the same sequence as W symbol for symbol, but its orbit visits kN distinct states before repeating, so its raw ledger depth saturates at kN rather than N. ■

Verification 9.11. Instantiated at N = 32 with k = 1, 3, 7: the three implementations produced byte-identical output sequences, and raw depths after 400 steps of 32, 96, and 224 respectively — exactly the factors 1, 3, 7. Minimizing each by output-equivalence returned 32, 32, 32.

Corollary 9.12. Ledger depth is a well-defined clock precisely on the minimal realization. On raw state it inflates by whatever redundancy the implementation carries, so two implementations of one computation would disagree about elapsed time.

Remark 9.13 (A falsifiable coefficient). The corrected identification supplies the coefficient its predecessor lacked. The claim is no longer a correlation between two monotone quantities, which is unfalsifiable, but the assertion that depth per output symbol on the quotient equals exactly one. This was confirmed for every substrate tested. A correlation between raw depth and step count would have been 1.0 by construction and could not have been evidence for anything.

10. Numerical Protocols

All computations use exact integer arithmetic; no floating-point value enters any structural decision. Prime factorizations are obtained by smallest-prime-factor sieve, and orders by prime-power accumulation over cycle lengths rather than by direct lcm, to avoid overflow.

10.1 Base objects. Three families recur. *Pure*: multiplication by an odd constant modulo 2^W, whose cycle lengths are all powers of 2, giving |S| = 1. *Rich*: uniform random permutations, with |S| ∈ {8, 9} at the widths used. *Extremal*: a single N-cycle. Widths W ∈ {6, 8, 10, 12, 14} are used, N = 2^W.

9.2 Surgery. A surgery is a 3-cycle on uniformly sampled distinct points, applied on the left. It lies in A_Ω and has the minimal non-identity support (Remark 0.5).

10.3 Continuity measurements (Section 4). A tunable cross generator was formed by composing a family member with a product of disjoint 3-cycles of controlled total support s. The family scaffold was held fixed across the sweep to remove the variance introduced by different multipliers having different gcd structure. At W = 12, with the scaffold fixed: O moved 10.00 → 19.65 → 23.50 → 28.65 → 34.55 → 37.46 at s = 0, 3, 6, 12, 24, 48, while F remained at exactly 4.00 throughout that range, first moving at $s = 192$ and reaching ≈ 1 only near $s = N$.

10.4 Span classification (Section 2). 3000 surgeries at $W = 6$ gave $\Delta K = -2$ in 1765/1765 span-3 cases, $\Delta K = 0$ in 1140/1140 span-2 cases, and $\Delta K \in \{0, +2\}$ split 48/47 in span-1 cases. The merge-length law held 231/231.

10.5 Compression law (Section 3). As stated in Verification 3.3: 1800/1800 exact.

10.6 Markov measurements (Section 6). 4000 uniform surgeries per transition law at $W = 10$; total variation as reported in Verification 7.2.

10.7 Drift measurements (Section 8). Trajectories of 500 steps from each of 4 starting spectra at $W = 12$ (2000 steps total), and of 300–500 steps from 3–4 starts at each of $W \in \{8, 10, 12, 14\}$. Bins with fewer than 25 observations were discarded. Zero crossings were located by linear interpolation between adjacent bin means.

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11. Ledger of Refuted Claims

Refuted claims are retained with the expansion that replaced each, on the principle that any narrowing of the theory must be accounted for.

R1. Opacity is inertia. Refuted by Remark 5.3 and, structurally, by Theorems 4 and 5. *Expansion:* Corollary 5.1, the proven independence of the two channels, with Remark 5.4 supplying the reason from (0.1).

R2. Coherence is binary. Refuted by the tunable cross generator of §9.3, which exhibits continuous degradation in O with the commutation deficit. *Expansion:* fragility is a property of a channel, not of a body; the two channels differ by more than two orders of magnitude in sensitivity scale.

R3. Support drives the order's response. Refuted by Verification 6.4 at constant support. *Expansion:* Theorems 1, 2 and 6 — cycle topology determines the partition update, and the arithmetic of the resulting lengths determines the order.

R4. Observables compose as an algebra. Refuted by Corrected test 7.5, which also inverted the registered prediction. *Expansion:* Conclusion 7.6, the flat coordinate system.

R5. The first two-port test. Its baseline was infeasible (Invalid test 7.4) and it returned four false positives. *Expansion:* the segment criterion of 7.5, and the general caution that on constrained state spaces any arithmetic baseline must be checked for membership in the feasible region before departures from it are interpreted.

R6. An external quadratic form reproducing lepton mass ratios. A form $F(p,q) = \alpha p^2 - \alpha pq + q^2$ with $\alpha = 4.08222$ and assignments (3,7), (3,8), (3,13) was proposed externally as a bridge to the settlement mass. Audit: α lies within 1.1×10^{-3} of 49/12, the value annihilating F(3,7), making the reference quantity a near-cancellation of magnitude 0.0134; a perturbation of 10^{-5} in α produces a 10% error in both ratios, so α must be specified to six significant figures; the only α-free content, the limit of the ratio-of-ratios, is 558/33 = 16.909 against the actual 16.817, an error of 0.55%; and an exhaustive search over coprime pairs with entries ≤ 12 found **392 triples** fitting both target ratios to within 0.5%, many to within 0.01%. One continuous parameter with three free integer pairs suffices to hit two targets. Independently, the settlement mass was measured against F(p,q) and did not match. *Expansion:* a boundary marker — external numerical coincidences are not admitted without derivation.

**R7.* *p\ ≈ 6.2 as a constant.* *Refuted by Proposition 8.4.* Expansion:* the scaling law (8.5) and the gain–loss model (8.4).

R8. The vertex/minimal/edge sandwich. It was claimed that |S| < |S_min| < |S × A| holds generally. Refuted by Theorem 9.7(v): at |A| = 2 the minimal realization is strictly smaller than |S| in roughly 45 percent of machines. The original measurement was taken at the single parameter point |S| = 8, |A| = 4, where the ordering happens to hold. *Expansion:* the stable invariant is |S_min| / |S × A| ~ 0.55, uniform across the parameter range; the minimum tracks the edge space and is unordered against the vertex.

R9. Time is ledger depth. Refuted by Proposition 9.10: implementations with identical output and raw depths in ratio 1 : 3 : 7 would carry three different clocks. *Expansion:* Corollary 9.12, depth is a clock on the minimal realization only, together with the falsifiable coefficient of Remark 9.13, one unit of depth per output symbol on the quotient.

R10. A stated bound reported as a value. The state information at N = 1024 was reported using a truncated factorial guard, yielding ≈1019 bits. The correct value is $\log_2(1024!) \approx$ **8769 bits**. The error understated the control-to-state gap by a factor of about 8, strengthening rather than weakening Remark 7.2.

12. Architecture and Open Problems

12.1 The surviving architecture

Three layers, of which only the middle evolves.

Verb layer. Surgery operators τ ∈ A_Ω of minimal support, acting by left composition. These are the only objects that act.

State layer. The cycle partition λ ∈ Λ_N. Its transition is closed-form (Theorem 3), determined by two applications of the transposition rule (Theorems 1, 2). Under label-independent surgery selection it is a sufficient statistic (Theorem 7, with Remark 7.1), so the dynamics is a Markov chain on Λ_N; the chain branches at ≥ 7.5 bits per step (Proposition 7.3).

Observable layer. Functionals of λ: O reads prime exponents through (0.1); F reads the multiplicity of 1; K, M, p read other coordinates. These do not evolve; they are read. They are provably independent (Corollary 5.1) and compositionally flat (Conclusion 7.6). Fed back, they act as steering ports of very low bandwidth relative to the state (Remark 7.2).

Realization. Closing an open machine adds statements rather than a fourth layer. Feedback forces additional state; the edge lift is the canonical machine-independent realization of it; the minimal realization is the quotient by output congruence (Theorem 9.7). Admissibility under feedback is governed by the interior operator nu, whose join failure is characterized by Theorem 9.3. Ledger depth is a clock on the quotient and nowhere else (Corollary 9.12).

The named mechanical quantities of the prior work are therefore downstream compositions. The inertial mass is O, and O is five steps from execution: surgery, partition, lengths, exponents, lcm, logarithm.

12.2 Open problems

O1 (Derivation of the drift coefficients). Open Problem 8.8. Derive G and c from the transition kernel, computing p*(W) rather than fitting (8.5).

O2 (Generality of the macroscopic law). D(p) closes for O. Whether F, K, or M admit analogous closed drift laws, and in which variables, is unmeasured. It is possible that O is distinguished because it is the only observable reading the spectrum's arithmetic.

O3 (Entropy as a function of state). Proposition 7.3 gives a lower bound at one partition. The dependence of H(λ_{t+1} | λ_t) on λ_t is unknown, and bears on the mixing time and hence on the rate of approach to the equilibrium of Proposition 8.4.

O4 (Intermediate compression). The orbit does not close at the level of λ; a scalar projection does. Whether a low-dimensional description intermediate between these exists — for instance in (p, K, M) jointly — is a direct question with a direct experiment behind it.

O5 (Hardware idempotence). The constraint operator ASSERT is idempotent in the algebra. Whether physical processors exhibit a measurable cost collapse on the second application of an invariant check — energy or cycle count for a repeated projection against a single one — is untested, and requires instrumented silicon rather than simulation.

O6 (Physical interpretation). That O behaves as an inertial cost and F as a persistence is a reading *within* this formalism. Connecting it to a configuration space, a connection, and an action functional has not been done, and one attempt to cross from the far side was refuted (R6). The mathematical results of Sections 2–8 do not depend on the physical reading and should not be presented as though the reading followed from them.

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13. Closing Remark

The programme's organizing claim is that verbs produce state and that nouns are functionals of state. This paper renders that claim formally. The verbs are the surgery operators; the state is the cycle partition, whose evolution is governed by Theorems 1–3 and whose sufficiency is Theorem 7; the nouns are the functionals of Definition 0.2, provably independent by Corollary 5.1 and provably flat by Conclusion 7.6. Complexity is not in the nouns, which are projections; not in the ports, which compose trivially; and not in the transition, which is closed-form. It is in the orbit of the partition under repeated surgery, and it is there that any further compression must be sought.

Appendix A. The Transition Algorithm

The algorithm of Theorem 3 in explicit form. The state is a list of cycles, each stored as an ordered sequence of points; loc maps a point to its (cycle index, position) pair.

```
INPUT   cycles C = [C_1, ..., C_k], surgery points a, b, c

OUTPUT  the partition lambda(tau P), tau = (a b c)

function TRANSPOSE(C, x, y):

  (i, p) <- loc[x];  (j, q) <- loc[y]

  if i = j:                        # SPLIT  (Theorem 1(i))

    L <- |C_i| ;  s <- (q - p) mod L

    R <- rotate C_i so that x is first

    replace C_i by  R[0..s-1]  and  R[s..L-1]

  else:                            # MERGE  (Theorem 1(ii))

    R1 <- rotate C_i so that x is first

    R2 <- rotate C_j so that y is first

    replace C_i, C_j by  R1 ++ R2

  return updated C

function TRANSITION(C, a, b, c):

  C  <- TRANSPOSE(C, a, b)              # first factor of (a c)(a b)

  C  <- TRANSPOSE(C, a, c)              # second factor

  return sorted multiset of |C_i|
```

Correctness is Theorems 1 and 2. The cost is dominated by the two rotations, giving O(L_max) where L_max is the largest affected cycle length; in particular it is independent of N except through that length, and no array of size N is ever written. By contrast, forming τP explicitly and decomposing it costs $\Theta(N)$.

Remark A.1. The asymmetry in cost is the computational shadow of the structural claim. Composing permutations treats the state as N labelled points; the transition law treats it as a partition with local repair. That the second suffices is Theorem 7.

Appendix B. Reproduction

Every figure quoted in this paper was produced by direct computation under the protocols of Section 9. The following table indexes the principal claims to their verifications.

Claim	Statement	Verification	Result		
Split/merge rule	Theorem 1	§10.4	231/231 merge lengths exact		
Span rules	Corollary 2.1	§10.4	1765/1765, 1140/1140, 48:47		
Compression law	Theorem 3	Verification 3.3	1800/1800 exact		
Order unbounded	Theorem 5	Verification 5.2	ratio = N/4, k = 4..16		
Purity implies fragility	Corollary 6.1	Verification 6.3	+6.20, +6.35 vs −2.56, −0.01		
Topology not support	Remark 6.2	Verification 6.4	corr +0.9933 vs +0.5300		
Sufficiency	Theorem 7	Verification 7.2	TV 0.2442 vs noise floor 0.2398		
Branching	Proposition 7.3	§10.6	972 successors, H ≥ 7.49 bits		
Ports independent	Experiment 7.1	§7.1	four distinct attractors		
Ports flat	Conclusion 7.6	Corrected test 7.5	d/	seg	≤ 0.35
Drift decomposition	Proposition 8.3	§9.7	gain flat, loss slope ≈ 1.3		

Claim	Statement	Verification	Result		
Equilibrium scaling	Proposition 8.4	§10.7	p* ≈ 0.94W – 2.8		
nu interior operator	Proposition 9.2	Verification 9.4	400/400 on four laws		
Straddle theorem	Theorem 9.3	Verification 9.4	400/400 agreement		
Disjunctive checkability	Corollary 9.5	Verification 9.6	3.60% of pairs		
Vertex insufficiency	Theorem 9.7(i)	Verification 9.8	6/300 machines		
Coarsest congruence unique	Theorem 9.7(iv)	Verification 9.8	200/200 exhaustive		
Depth not invariant	Proposition 9.10	Verification 9.11	32:96:224 -> 32:32:32		